SECONDS, LIFETIMES AND ORDERS OF MAGNITUDE ON ONE PAGE

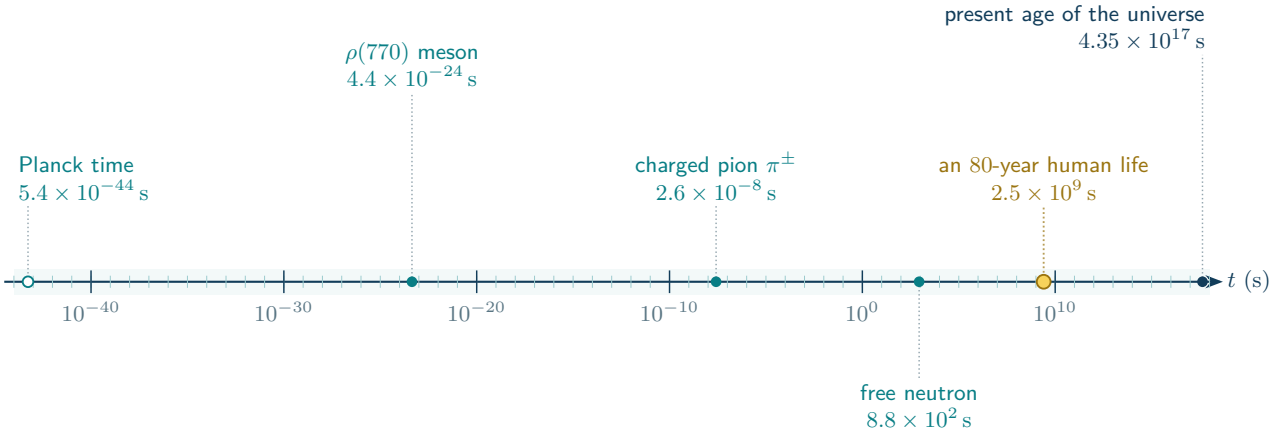
How long is a lifetime?

A logarithmic ruler from the rho meson to a human life and the age of the universe.

1 / PUT EVERY DURATION ON THE SAME AXIS

One ruler across about 61 orders of magnitude

$$x = \log_{10}\left(\frac{t}{1\,\mathrm{s}}\right), \qquad x_2 - x_1 = \log_{10}\left(\frac{t_2}{t_1}\right).$$



Each small tick is one **decade**: moving one tick right multiplies the duration by 10. The positions compare lengths of time.

2 / START WITH A FAMILIAR LIFESPAN

A human life is a few billion seconds

Using a 365.25-day year,

$$1\,\mathrm{yr} = 365.25 \times 24 \times 3600\,\mathrm{s} \approx 3.156 \times 10^{7}\,\mathrm{s}.$$

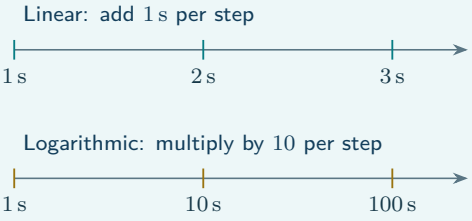
The mnemonic $1\,\mathrm{yr} \approx \pi \times 10^{7}\,\mathrm{s}$ is within about 0.5%. An illustrative human lifespan is therefore

$$T_H \;=\; 80\,\mathrm{yr} \;\approx\; 2.5 \times 10^{9}\,\mathrm{s}$$

Its log-position is $x_H \approx 9.40$, between 10^9 and 10^{10} seconds. Negative x means a positive duration shorter than one second.

3 / READ EQUAL SPACING CORRECTLY

A logarithmic midpoint is geometric



Halfway along a log interval means averaging the logarithms:

$$t_{\mathrm{mid}} = \sqrt{t_1 t_2}.$$

Between 1 s and 100 s, the midpoint is 10 s.

4 / GAUGING A LIFETIME IN RHO UNITS

How many rho lifetimes fit in ours?

Use $\tau_\rho \approx 4.4 \times 10^{-24}\,\mathrm{s}$ as a reference interval. The number of these intervals in an 80-year life is

$$N_H = \frac{T_H}{\tau_\rho} \approx \frac{2.5 \times 10^{9}}{4.4 \times 10^{-24}} \approx 5.7 \times 10^{32}.$$

The gap on the log ruler is therefore

$$\Delta x \;=\; \log_{10} N_H \;\approx\; 32.8 \quad \text{(about 33 decades)}$$

Even two tiny particle lifetimes can be enormously different:

$$\frac{\tau_{\pi^\pm}}{\tau_\rho} \approx 5.9 \times 10^{15},$$

about 16 orders of magnitude between the charged pion and the rho. Here a count means intervals of length τ_ρ.

5 / FROM PLANCK TIME TO COSMIC AGE

About 61 decades fit on one page

Compare the theoretical Planck time $t_P \approx 5.4 \times 10^{-44}\,\mathrm{s}$ with the universe's present age, about 13.8 billion years:

$$t_U \approx 4.35 \times 10^{17}\,\mathrm{s}.$$

$$\frac{t_U}{t_P} \;\approx\; 8.1 \times 10^{60}, \qquad \Delta x \;\approx\; 60.9$$

The Sun's main-sequence duration, roughly 10^{10} years or $3.2 \times 10^{17}\,\mathrm{s}$, lies close to the cosmic-age mark on this ruler.

Changing to rho units shifts the zero:

$$x_\rho = \log_{10}\left(\frac{t}{\tau_\rho}\right) \approx x + 23.36.$$

The time ratios and their log-separations stay the same.

Conventions. All $t > 0$. Particle entries are approximate rest-frame mean lifetimes; Planck time is a theoretical scale, and the cosmic entry is an age. The human lifespan is a chosen example.

Origin. Lee McCulloch-James (2022), *Gauging a lifetime on skid rho*. **Values.** PDG mesons, PDG neutron, NIST/CODATA, NASA cosmos, NASA stars.

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